

A short applied data-analysis case study

AUTHOR
Birk Tyssebotn

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Background

In late July 2026, the United States imposed an additional 12.5% tariff on Norwegian goods, replacing the temporary 10% global tariff that expired on 24 July 2026. The U.S. justified the higher rate for Norway under a Section 301 investigation into forced labour in global supply chains – a justification the Norwegian government has publicly rejected.¹

Worth noting up front: the OECD’s 2026 Economic Survey of Norway characterizes the country as relatively insulated from U.S. trade policy overall, since the U.S. takes well under 4% of Norway’s total exports.² The numbers below sit inside that broader picture – aggregate exposure is low, but that headline figure can still hide industries where the U.S. is a much larger share of *that industry’s own* exports (see “How dependent is each industry on the U.S. market?” below).

The tariff is barely a month old at the time of writing, and SSB’s trade statistics lag by about a month, so there isn’t yet a meaningful window of *post-tariff* data to analyse – one partial month tells you nothing statistically. Rather than force a before/after comparison the data can’t support, this case study answers a question the data *can* answer well:

1. **How much**, and in which industries, does Norway export to the U.S., and how dependent is each industry on that specific market?
2. **How stable is that trade normally** – seasonally and otherwise – before any policy shock, so a future change can be judged against a real baseline instead of a flat one?
3. **How much is actually at stake**: if the 12.5% tariff applied to recent export volumes as-is, what would that cost, by industry?

Data

Source: Statistics Norway (SSB), StatBank Table [08806 – External trade in goods, by commodity group, country, month and imports/exports](#). Data are licensed under [CC BY 4.0](#).

Industries are classified using the UN’s [Standard International Trade Classification \(SITC\)](#), 1-digit level – broad enough to read at a glance, granular enough to see which sectors are actually exposed to the tariff.

Three extracts, all with the same commodity/exports settings, differing only in country and time span:

File	Country	Months	Used for
<code>exports_us_monthly.xlsx</code>	United States	Jan 2025 – present	Main totals, share, exposure
<code>exports_total_monthly.xlsx</code>	All countries	Jan 2025 – present	US share of Norway’s total exports
<code>exports_us_monthly_long.xlsx</code>	United States	Jan 2018 – present	Seasonal baseline

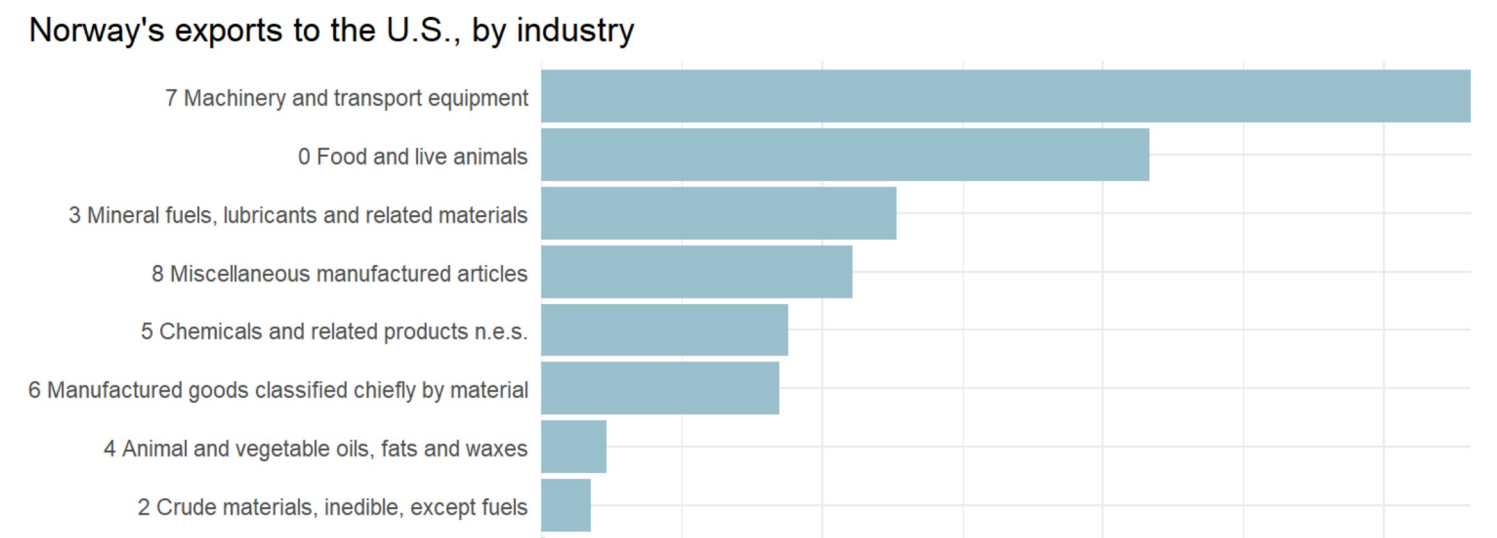
Tidied in `scripts/00_helpers.R` (shared parsing logic) and `scripts/01_wrangle_exports.R`, `02_wrangle_share_of_total.R`, `03_wrangle_seasonal.R`.

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How much does each industry export to the U.S.?

Total exports and export share, most recent 12 months, sorted by value.

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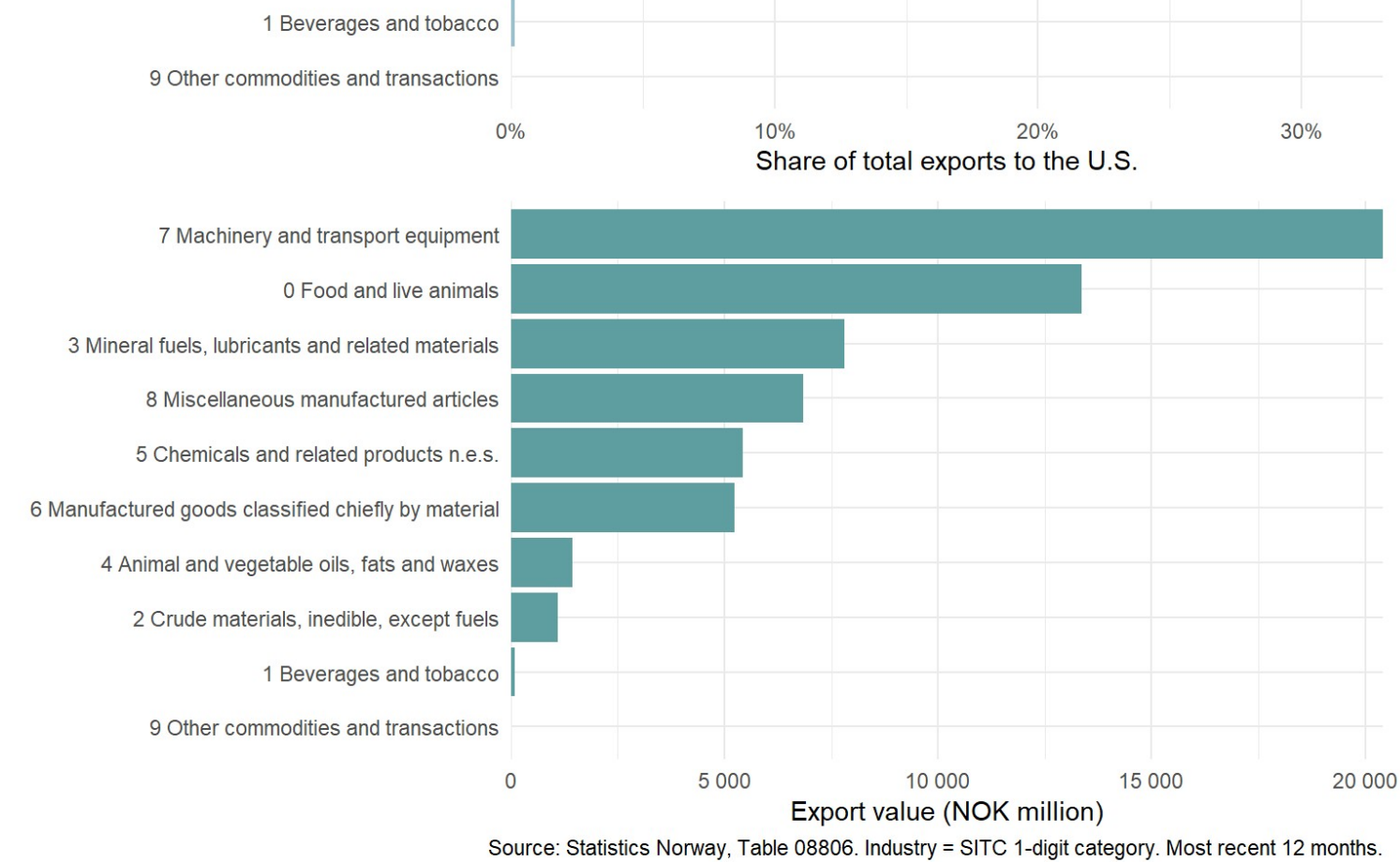


Figure 1: Norway's exports to the U.S. by industry: value and share, most recent 12 months

How dependent is each industry on the U.S. market?

A different question from the one above: not “what share of US-bound exports is this industry,” but “what share of *this industry's total exports* goes to the US specifically.” An industry can be small in the US market overall but still highly exposed, if the US is one of its only buyers.

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How exposed is each industry to the U.S. market specifically?

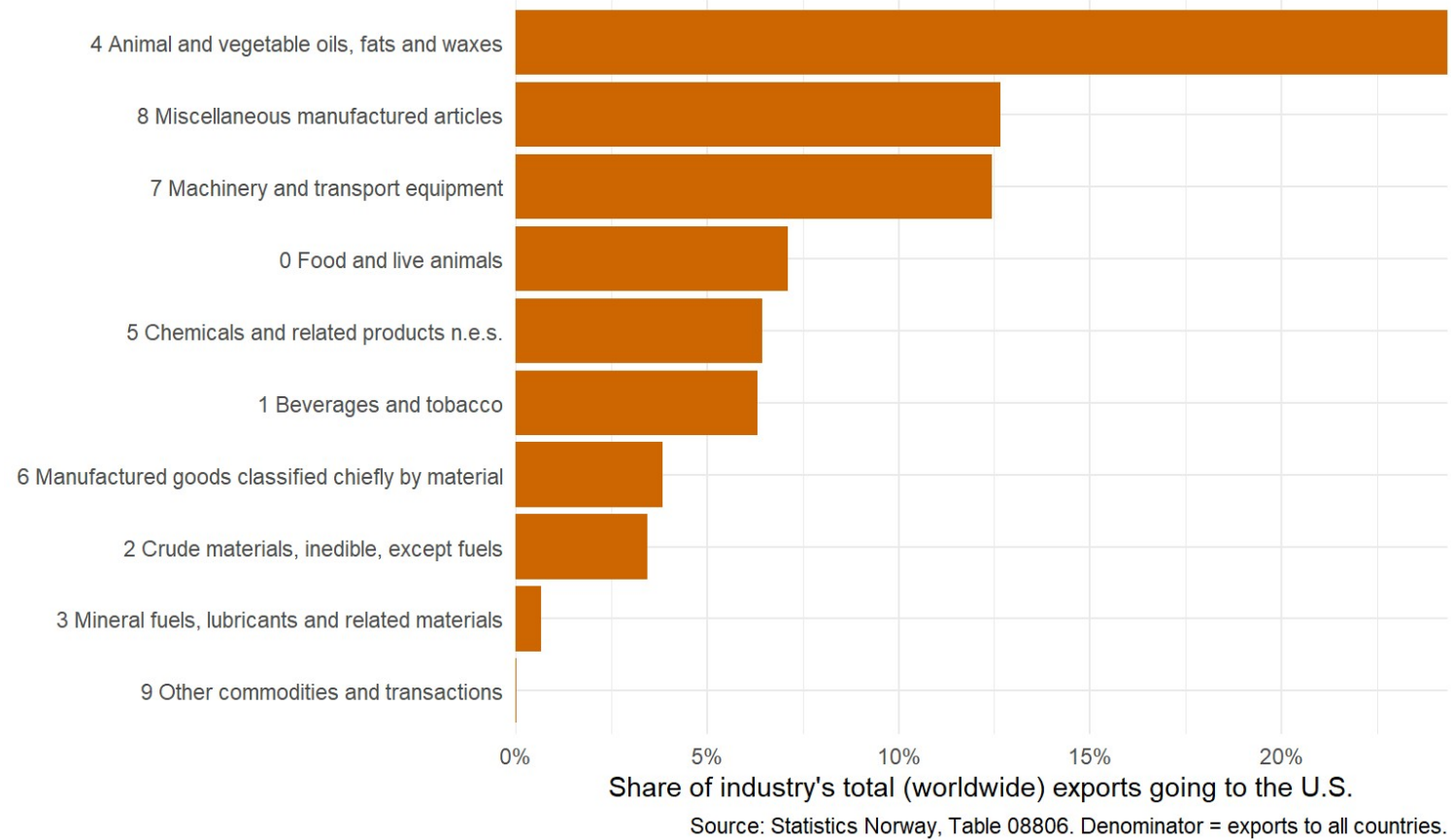


Figure 2: Share of each industry's total (worldwide) exports that goes to the U.S., most recent 12 months

How volatile is this trade normally?

Before drawing any conclusion from a future dip or spike, a baseline for normal month-to-month variation is needed. Some industries (mineral fuels, for instance) move on large one-off shipments even with no policy change involved; others (food) are steadier.

The chart also marks two dates: **2 April 2025** (“Liberation Day”), when the U.S. first announced broad reciprocal tariffs on nearly every country including Norway, and **24 July 2026**, the current 12.5% tariff this case study is about. The first gives a real precedent for what a genuine tariff shock looks like in this exact data – a useful reference point for judging whether the second produces anything comparable once more months are published.

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Norway's monthly exports to the U.S., by industry

Dashed lines: 2 Apr 2025 (Liberation Day tariffs) and 24 Jul 2026 (current tariff). Data does not yet cover a meaningful window after the second



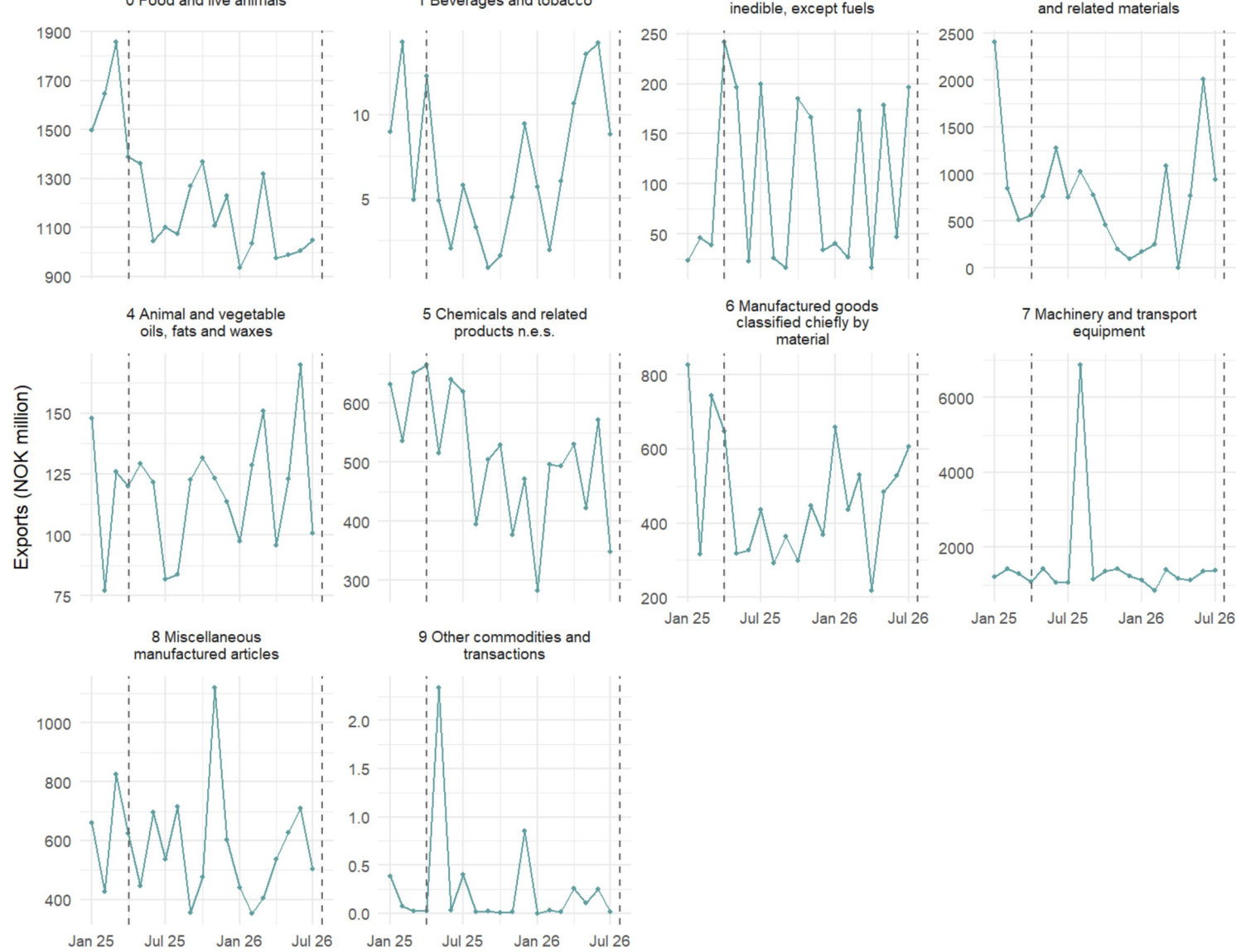


Figure 3: Monthly exports to the U.S. by industry, Jan 2025 to present

A seasonally-adjusted baseline

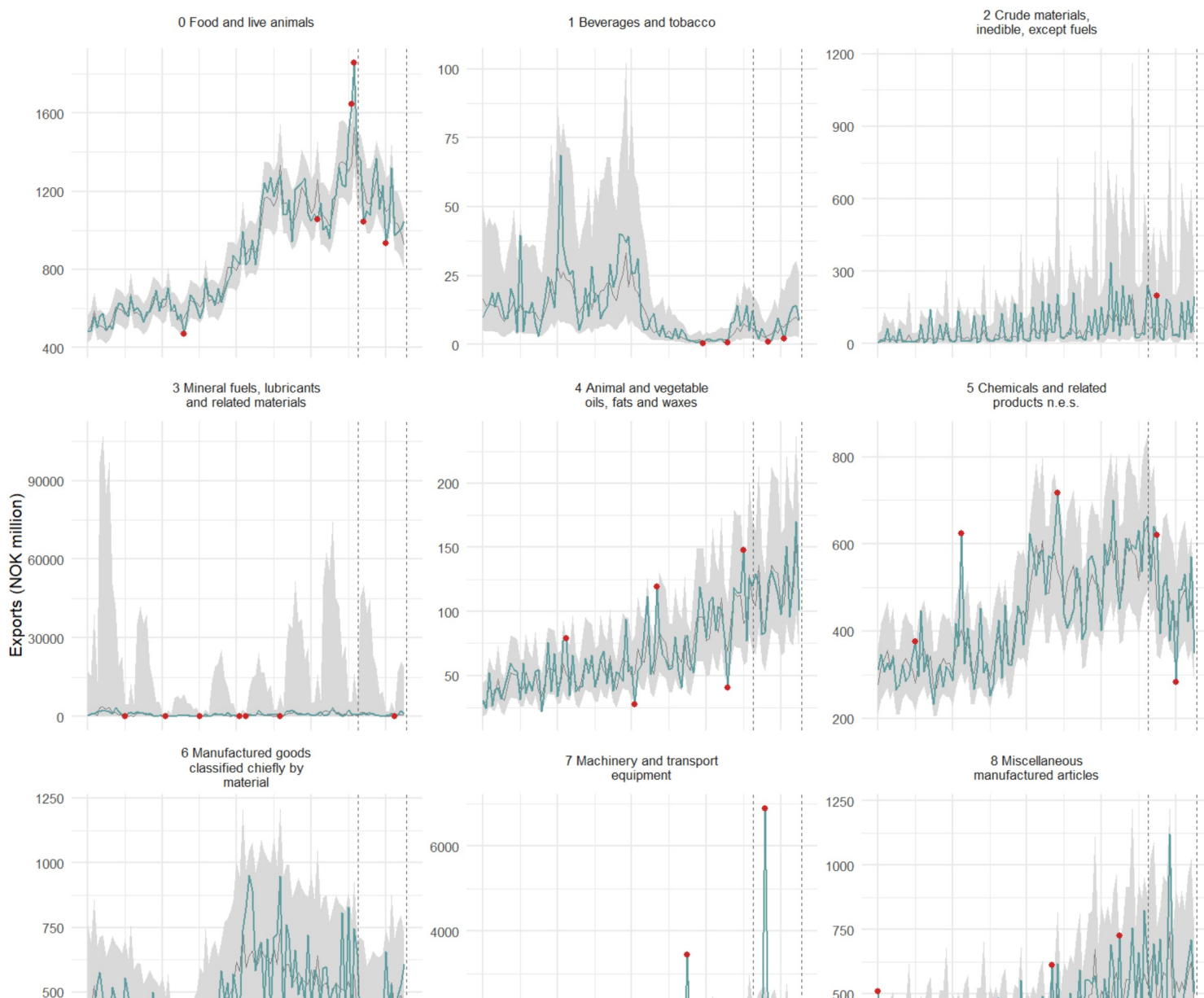
The chart above shows raw variation, which mixes three things together: a long-run trend, a repeating within-year seasonal pattern, and genuine one-off noise. To say a given month is “unusual” requires separating those out first.

Using nine years of history (2018 – present; too few years wouldn’t support a reliable seasonal estimate), each industry’s monthly exports are decomposed via STL (Seasonal-Trend decomposition using Loess) on the log scale into trend, seasonal, and remainder components. The remainder’s historical standard deviation defines “normal”: a month is flagged as **out of the normal range** if its remainder exceeds **2 standard deviations**, i.e. bigger than seasonality and typical random variation alone would usually produce. “9 Other commodities and transactions” is excluded – too small and irregular (multiple months recorded as zero or suppressed) for a seasonal pattern to mean anything.

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Actual exports vs. the seasonally-expected range

Grey band = normal range (+2 SD of remainder). Red points = out of that range. Dashed lines: Apr 2025 and Jul 2026 tariff dates.



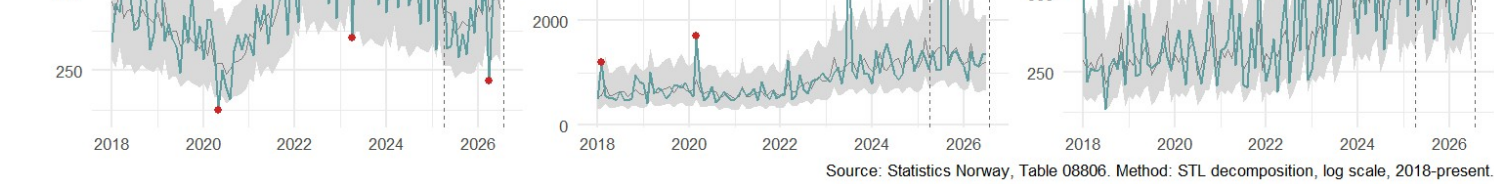


Figure 4: Actual vs. seasonally expected exports, 2018-present, with out-of-range months flagged

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Table 1: Status of the most recent available month against each industry’s seasonal baseline

Industry	Actual (NOK mill.)	Seasonally expected (NOK mill.)	Out of normal range?
0 Food and live animals	1048	926	No
1 Beverages and tobacco	9	8	No
2 Crude materials, inedible, except fuels	197	47	No
3 Mineral fuels, lubricants and related materials	943	628	No
4 Animal and vegetable oils, fats and waxes	101	107	No
5 Chemicals and related products n.e.s.	349	366	No
6 Manufactured goods classified chiefly by material	607	443	No
7 Machinery and transport equipment	1371	1196	No
8 Miscellaneous manufactured articles	503	439	No

What’s at stake: a naive tariff-exposure estimate

If the 12.5% additional tariff applied to the last 12 months of exports exactly as they were, this is what it would have cost, by industry. This is not a forecast – exporters adjust prices, currencies move, volumes shift in response – but it puts a concrete number on exposure, which the totals chart alone does not.

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What the 12.5% tariff would cost, by industry

Naive estimate: 12.5% x trailing 12-month export value. Assumes no change in price, volume, or currency.

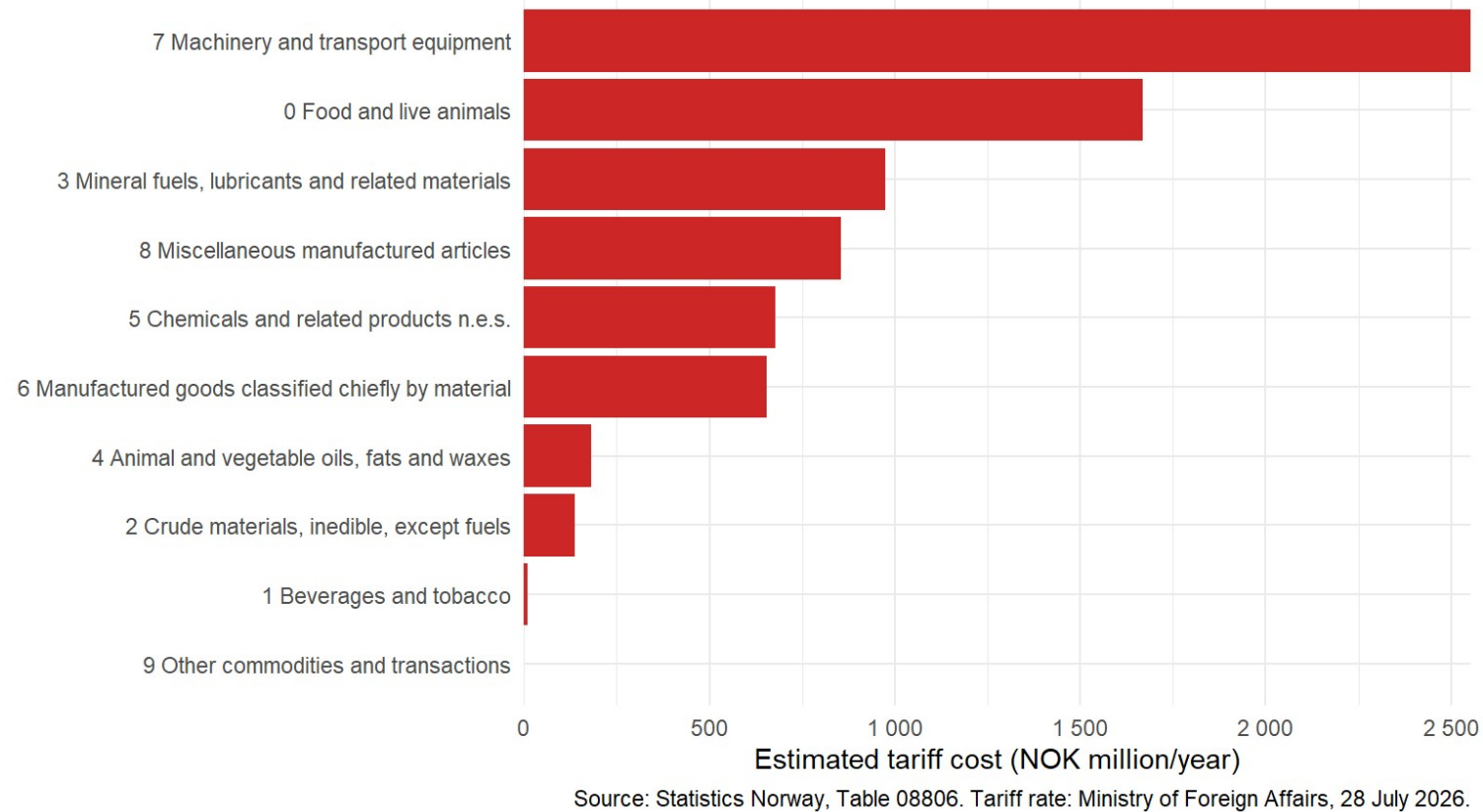


Figure 5: Estimated annual tariff cost if applied to the last 12 months of exports, by industry

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Table 2: Exposure estimate by industry, most recent 12 months

Industry	Exports (NOK mill.)	Share of total	Est. tariff cost (NOK mill.)
7 Machinery and transport equipment	20421	33.1%	2553
0 Food and live animals	13356	21.6%	1669
3 Mineral fuels, lubricants and related materials	7805	12.6%	976
8 Miscellaneous manufactured articles	6845	11.1%	856
5 Chemicals and related products n.e.s.	5422	8.8%	678
6 Manufactured goods classified chiefly by material	5231	8.5%	654
4 Animal and vegetable oils, fats and waxes	1441	2.3%	180
2 Crude materials, inedible, except fuels	1106	1.8%	138
1 Beverages and tobacco	81	0.1%	10
9 Other commodities and transactions	2	0.0%	0

Takeaways

Write this section once you’ve looked at your own rendered plots – don’t let me write your conclusions. A few honest starting points based on the structure of the data itself:

- *Machinery and transport equipment, and food, are the two largest categories – together they likely account for a large share of total tariff exposure, simply because they’re the largest exports, not because they’re specially targeted.*
- *An industry can be small within the US market (fig-dependence low rank) but still highly exposed if it has few other buyers – check whether the two rankings (share-of-US-market vs. share-of-own-exports) actually agree.*
- *tbl-seasonal-latest gives a direct, falsifiable answer to “has anything already moved beyond normal”: read it plainly rather than reading intent into it – one industry crossing the 2-SD line is exactly what you’d expect by chance roughly 1 month in 20, tariff or not. It’s a flag to watch, not a verdict.*
- *The exposure estimate is a ceiling, not a prediction: real-world responses (price pass-through, substitution to other markets, exchange rate movements) will change the actual impact. Say that explicitly rather than implying the bars are what will happen.*

Notes on methodology

- **Units:** NOK million, current prices (not inflation-adjusted).
- **Industry classification:** SITC 1-digit – broad by design, for readability. A finer breakdown (2- or 3-digit SITC) would let you check whether, say, farmed salmon specifically drives the seafood category.
- **“9 Other commodities and transactions”** is excluded from the seasonal analysis: several months are recorded as zero or suppressed in the source data, and the category is under 0.03% of total exports (see the exposure table) – too sparse and irregular for a seasonal pattern to be meaningful.
- **Seasonal baseline method:** STL decomposition on log-transformed monthly exports, `s.window = "periodic"` (assumes a stable seasonal pattern across the 2018-present window rather than one that drifts over time). The 2018-2021 portion includes the COVID trade disruption, which will inflate the estimated “normal” variation for some industries – a stricter baseline would exclude those years, at the cost of a shorter, noisier history to estimate from. Worth trying both and comparing if this section needs to hold up to scrutiny.
- **Why no before/after comparison of the current tariff:** it took effect 24 July 2026; at the time of writing, SSB’s most recent published month is July 2026, only partially post-tariff. A credible comparison needs several full months on each side. The seasonally-adjusted baseline above is what that future comparison should be checked against, rather than a flat pre/post average.
- **Tariff-exposure estimate is naive by design:** it holds trade volumes, prices, and exchange rates fixed and simply applies the tariff rate to historical exports. It’s a “what’s on the table” figure, not an economic forecast – real elasticities would move the actual number.
- **Reproducibility:** re-running `scripts/01_wrangle_exports.R`, `02_wrangle_share_of_total.R`, and `03_wrangle_seasonal.R` after downloading updated extracts from SSB regenerates every processed CSV and every figure in this document.

Footnotes

1. Norwegian Ministry of Foreign Affairs, “The US introduces a new tariff rate on goods from Norway,” 28 July 2026, <https://www.regjeringen.no/en/whats-new/the-us-introduces-a-new-tariff-rate-on-goods-from-norway/id3169293/>. ↩
2. OECD, *OECD Economic Surveys: Norway 2026*, “Preparing for long-term challenges,” https://www.oecd.org/en/publications/oecd-economic-surveys-norway-2026_5cc02644-en/full-report/preparing-for-long-term-challenges_b3a8411e.html. ↩